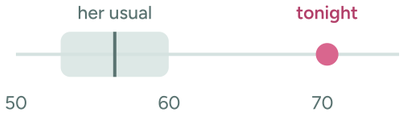
How the twin decides, step by step

Worked on Meera Iyer (P-0007) on 12 October 2026. Every number below is computed by the trained model on the synthetic cohort.

1 Her usual

Personal baseline

Tonight is compared with her own last 28 nights, not with other people.



Resting heart rate, bpm

56.5 usual (28-night median)

70.3 tonight

+13.8 bpm

above her usual. Is that worrying?

Same idea for HRV (60 nights) and home BP (28 days).

2 How her body responds

Personal response slopes

Learned from 109 calm days: what a short night or a lazy day does to her.

1 hour less sleep
heart rate next night
+1.4 bpm most people +1.0 bpm

1,000 fewer steps
heart rate next night
+0.4 bpm most people +0.3 bpm

1,000 more steps
BP next morning
-1.3 mmHg most people -1.3 mmHg

1 hour more sleep
HRV next night
+3.8 % most people +4.4 %

A regression of tomorrow on today, one person at a time, corrected for drift.

3 Borrow from the cohort

Empirical Bayes pooling

A few months of one person's data is noisy. So each estimate is pulled toward the cohort, by exactly as much as it is unsure.



Sleep to heart rate, bpm per hour

Weight on her own data

= spread² / (spread² + noise²)

= 0.43² / (0.43² + 0.37²) = 57%

Twin = 57% × (-1.69) + 43% × (-1.03) = -1.41

New users start at the cohort value; the more calm days, the more it is theirs.

4 Explained or not?

Twin-adjusted signal

Subtract what her own habits explain. What is left is the warning signal.

Last night 1.4 h less sleep than usual, yesterday 1,498 more steps.



Explained = her slopes × her habits

= (-1.38) × (-1.38) + (-0.43) × (1.50)

= +1.26 bpm

Unexplained = 13.8 - 1.3 = 12.6 bpm

Home BP: +16 mmHg her activity cannot explain.

Slopes use only days before tonight, so no future data leaks into the score.

5 Fuse, calibrate, decide

Calibrated early warning

LightGBM joins the unexplained signals, BP trend and eight health-record facts.

24% chance of a strain episode in 72 hours

likely 17% to 28%, her usual 0.7%

Score 63 of 100: Urgent

Why, in her doctor's words

- Home BP trending +2.1 mmHg/day over 7 days
- Home BP +16 mmHg beyond what activity explains
- Heart rate +13 bpm beyond what sleep and activity explain

Does it work? Mean of three patient splits, on people the model never saw

0.876
AUROC with steps 1 to 5

0.851
AUROC without steps 2 to 4 (no twin)

70%
of episodes caught at 0.045 false alerts per patient-day

r 0.59 to 0.71
match between the twin's slopes and each person's hidden true slopes